GearShare

Peer-to-Peer Gear Borrowing & Sharing Platform

Project Title

GearShare – Peer-to-Peer Gear Borrowing & Sharing Platform

Abstract

GearShare is a full-stack web application designed to provide a convenient platform for people to share and borrow equipment that they do not use regularly. Users can register on the platform, list their available gear with details such as category, description, and availability, and search for equipment based on their requirements. A user can submit a borrowing request for a specific period, while the owner can review and approve or reject the request. The system manages the complete borrowing lifecycle, including availability checking, request approval, active borrowing, return tracking, and borrowing history. Reviews and ratings can also be provided after a completed borrowing transaction to improve trust between users. The application will be developed using Java Spring Boot for the backend, Spring Data JPA and Hibernate for database interaction, MySQL for data storage, and React for the frontend. The project demonstrates practical concepts such as authentication, role-based access, REST APIs, relational database management, and conflict-free gear booking.

Project Workflow

Step	Process
1	User Registration / Login → User creates an account and securely logs into GearShare.
2	List Gear → Owner adds equipment with name, category, description, condition, and availability.
3	Search Gear → Borrower searches and filters available equipment according to requirements.
4	Check Availability → System verifies whether the selected gear is available for the requested dates.
5	Borrow Request → Borrower submits a request specifying the required borrowing period.
6	Owner Decision → Gear owner reviews the request and approves or rejects it.
7	Borrowing → After approval, the gear status changes to borrowed/reserved for the selected period.
8	Return → Borrower returns the gear and the owner records the return.
9	Review & Rating → Users can rate and review the completed borrowing transaction.
10	History → Users can view their listed gear, requests, active borrowings, and completed transactions.

Technology Stack

Frontend: React

Backend: Java, Spring Boot, Spring Security, REST APIs

Database: MySQL